

Homotopy Distillation of a Transformer into a Predictive-Coding Network

Faithful re-expression of GPT-2 (124M) with local, error-based learning

Technical report, work in progress

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Abstract

Predictive coding (PC) is a biologically motivated alternative to backpropagation in which learning is driven by *local* prediction errors rather than a global backward pass. A standing obstacle is scale: PC is rarely demonstrated beyond small networks, and openly documented PC transformers remain at the order of 10^5 parameters. We re-express a full-scale transformer, GPT-2 at 124M parameters, three orders of magnitude beyond that scale, as a predictive-coding network by *homotopy distillation*: a pretrained transformer is continuously deformed into an error-parameterised PC network along a schedule $\tau = \eta T$, from a backpropagation anchor at $T=1$ to a fully relaxed PC regime at terminal $\tau=6.4$. Three results are established to date. (i) The distillation is near-lossless and reproducible: across five independent runs the student’s next-token distribution tracks the teacher to a prompt KL of $\sim 10^{-5}$ nats at *every* milestone, nearly four thousand times inside the divergence-abort threshold, with held-out perplexity within 0.4%; both are $\varepsilon=0$ feedforward evaluations with no teacher present. A completed 5×10^7 -token production run on an OpenWebText shard extends this by an order of magnitude, worst milestone KL 3.1×10^{-5} . (ii) Along the homotopy the learning signal deforms as designed: the per-block PC weight update, probed teacher-free against plain backpropagation, grows from the anchor identity η at $T=1$ to full backpropagation magnitude, its direction staying aligned at a per-block cosine above 0.998 throughout. (iii) At 124M the PC error field is strongly concentrated: the top 5% of error cells carry $\approx 90\%$ of the squared update mass, the frontier sits in early and middle blocks rather than late ones, and both properties survive a teacher-free cross-entropy settle. This concentration is the lever for compressing the settle, the dominant cost of PC at scale. In companion work the same transformer’s full forward pass runs natively inside the MORK symbolic knowledge store, matched to the reference implementation to 6×10^{-6} relative error; predictive coding is the learning mode designed for that substrate, and the two results are halves of one programme. We close with the reproducibility methodology, an explicit statement of what is *not* yet shown, and the planned experiments.

1 Motivation and scope

Predictive coding learns by settling a network of state and error units to a low-energy configuration and then updating weights from purely local error signals; under standard assumptions its updates approximate those of backpropagation on arbitrary computation graphs [1, 2, 3]. This locality is attractive for two properties backpropagation does not offer: learning can proceed *online and in place* without a global backward pass, and the resulting dynamics are a candidate substrate for continual learning without catastrophic forgetting. The obstacle has always been scale. Public demonstrations of PC on transformer-shaped models remain at toy sizes (order 10^5 parameters), and it is an open question whether the PC parameterisation is even *expressive* enough to hold a modern transformer at realistic width and depth.

This work attacks that question directly, but by a deliberately conservative route. Rather than training a transformer from scratch under PC, which is compute-prohibitive for reasons dis-

cussed in §5, we ask whether an already-trained transformer can be *re-expressed* as a predictive-coding network without loss of function, at 124M parameters. A positive answer establishes expressivity at scale and yields the substrate on which the continual-learning properties that motivate PC can then be tested.

Contribution, stated precisely. This report claims: (a) a method, homotopy distillation, that deforms a pretrained transformer into an error-parameterised PC network; (b) an affirmative answer, at 124M, to the expressivity question: the PC parameterisation holds a modern transformer, near-losslessly and reproducibly, with local updates; and (c) a measurement of the PC error-mass distribution not previously available at this scale. It *does not* claim that PC trains a transformer from scratch, nor that PC outperforms backpropagation; those are separate questions, one of which (the continual-learning advantage) is a designed but not-yet-run experiment described in §7.

2 Method

The teacher is a frozen, pretrained GPT-2 (124M) [6]. The student is initialised from the teacher and trained under an error-based (ePC-style) parameterisation in which the prediction errors are the only free leaves and weight updates are computed locally from them [4, 5]. Training minimises a temperature-2 knowledge-distillation KL to the teacher.

2.1 The energy and the two updates

Write f_ℓ , $\ell = 1, \dots, L$ with $L=12$, for the transformer blocks and x_0 for the embedding output. The student’s forward pass carries an explicit error tensor at every block output:

$$x_\ell = f_\ell(x_{\ell-1}) + \varepsilon_\ell, \quad \hat{z} = \text{head}(\text{LN}_f(x_L)), \quad (1)$$

so the errors $\varepsilon = (\varepsilon_1, \dots, \varepsilon_L)$ are the only free variables of inference. At sequence length 512 this gives $12 \times 512 \times 768 = 4.72\text{M}$ error cells per sample, the cells measured in §3.3. Inference relaxes the energy

$$E(\varepsilon) = \sum_{\ell=1}^L \frac{1}{2} \|\varepsilon_\ell\|^2 + \mathcal{L}(\hat{z}(\varepsilon)), \quad (2)$$

where $\mathcal{L}$ is the distillation KL to the frozen teacher, by T steps of gradient descent $\varepsilon \leftarrow \varepsilon - \eta \nabla_\varepsilon E$ from $\varepsilon^{(0)}=0$, with the weights frozen (asserted at runtime, not assumed). The weights then descend the local objective

$$E_W = \sum_{\ell=1}^L \frac{1}{2} \|f_\ell(\bar{x}_{\ell-1}; W_\ell) - \bar{x}_\ell\|^2 + \mathcal{L}(\text{head}(\text{LN}_f(\bar{x}_L))), \quad (3)$$

where bars denote settled states held constant. Block ℓ regresses onto its own settled target, with $\nabla_{W_\ell} E_W = -\varepsilon_\ell \partial f_\ell / \partial W_\ell$ at the settled point, and the output head trains from the task loss on a constant trunk: the update is local by construction, not by approximation. The optimiser is AdamW on $\nabla_W E_W$.

The anchor at $T=1$ is an identity, not an analogy. At $\varepsilon = 0$ the forward (1) is exactly the vanilla network, so $\nabla_{\varepsilon_\ell} E = \partial \mathcal{L} / \partial x_\ell$, the backpropagation adjoint δ_ℓ . One relaxation step sets $\varepsilon_\ell^{(1)} = -\eta \delta_\ell$, so the block-weight gradient of (3) equals η times the backpropagation gradient to first order in η , and the head gradient is the task gradient itself; this is the discrete form of the PC-approximates-backprop correspondence [2, 5]. Increasing T moves inference from this anchor toward the PC equilibrium.

2.2 The homotopy schedule

The relaxation depth T follows a geometric schedule $T \in \{1, 2, 4, \dots\}$, and the effective relaxation time is $\tau = \eta T$ with step size η fixed within a run; the schedule runs from the $T=1$ anchor to terminal $\tau=6.4$, a fully relaxed PC regime: eight rungs at $\eta = 0.05$, seven at $\eta = 0.10$. A stage advances only when an energy-convergence hold criterion passes, and its τ step subdivides when it does not. Increasing T deforms the network continuously from backpropagation toward PC equilibrium, so that at no point does training leave a regime in which the student is a faithful copy of the teacher. Monitoring uses scale-invariant and rank statistics throughout (an energy-convergence ratio and trailing-median baselines) rather than absolute thresholds, for reasons that §4 makes concrete.

Each run of the replicate family processes 5.0×10^6 tokens at sequence length 512. The 5.0×10^7 -token production run has since completed; its fidelity is reported alongside the replicate results below.

3 Results established to date

3.1 The re-expression is near-lossless and reproducible

Figure 1 shows the student–teacher prompt KL across the anneal for five independent runs (the development run plus a four-cell replicate matrix spanning two random seeds and two micro-batch shapes). The worst milestone KL across all five runs is 1.3×10^{-5} nats, roughly 3.9×10^3 times inside the divergence-abort threshold, and held-out perplexity stays within 0.4% of the teacher throughout. Both quantities are evaluated on the student’s plain feedforward pass, $\varepsilon = 0$ with no settle, and the teacher is absent from the evaluation: these are deployment-condition numbers. The teacher enters training only through the settle objective $\mathcal{L}$ of §2. The divergence-abort guard never fired in any run, so no run was excluded from this set. The fidelity holds at every stage of the deformation, not only at its endpoint, and reproduces across seeds and batch shapes. This answers the expressivity question of §1 in the affirmative at 124M: the PC parameterisation holds a modern transformer.

The result now extends by an order of magnitude. The 5.0×10^7 -token production run trains at error step $\eta = 0.10$ with micro-batch 10 in fp32, a configuration first validated by its own replicate cell at pilot scale, over a seven-rung schedule to the same terminal $\tau = 6.4$; the training data is a pre-tokenised OpenWebText shard, and the run completed in 35.7 hours on a single A100. Fidelity holds at ten times the tokens: the worst milestone KL across its fifty-one milestones is 3.1×10^{-5} nats, sixteen hundred times inside the abort threshold, the final milestone sits at 5.8×10^{-6} , and held-out perplexity stays within 0.58% of the teacher throughout, the terminal deviation in the student’s favour. The run required no energy-convergence holds and no stage subdivisions end to end; its gradient-space path is taken up in §3.2.

3.2 The learning signal grows while its direction is held

Fidelity alone could be trivial if the relaxation never left the backpropagation anchor. Figure 2 shows what the homotopy does to the learning signal, measured teacher-free: at every milestone a probe computes, per block, the weight gradient of the settled local objective (3) and the weight gradient of plain backpropagation, both under a cross-entropy loss on data. At the anchor the norm ratio between the two is 0.05, equal to η , exactly as the $T=1$ identity of §2.1 predicts. By terminal τ the ratio reaches 0.98, full backpropagation magnitude, while the per- τ gain, the ratio divided by τ , falls from 1 to 0.15: the settle is genuinely deep, no longer a scaled single step. Through the whole deformation the minimum per-block cosine between the PC and backpropagation gradients never falls below 0.998. The learning signal leaves the anchor regime in magnitude while never leaving backpropagation’s direction at any depth; this is the PC-approximates-backprop correspondence [2] observed block-resolved at 124M along a continuous path. The 5×10^7 production run retraces this path at doubled error step: the ratio opens at

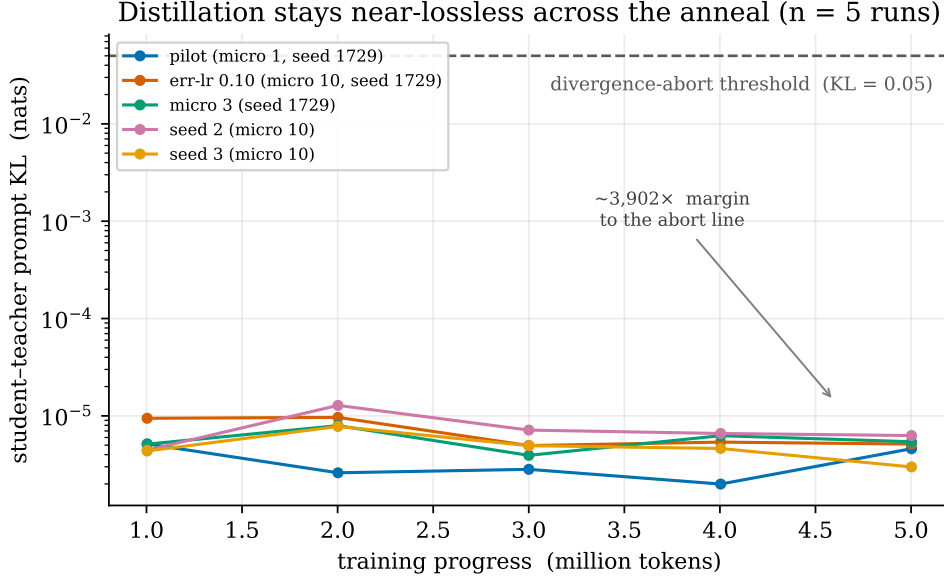


Figure 1: Distillation fidelity. Student-teacher prompt KL (log scale) across the anneal for five independent runs. All runs remain near 10^{-5} nats, about four thousand times inside the abort threshold (dashed); the KL is measured between plain feedforward passes of student and teacher ($\varepsilon=0$, no settle). The 5×10^7 production run holds the same band, worst milestone 3.1×10^{-5} (§3.1). This is a re-expression of a pretrained model, not from-scratch training.

the anchor identity again, 0.1000 against $\eta = 0.10$, saturates to 0.98, and the per-block cosine floor is 0.9986.

3.3 The error field is strongly concentrated at scale

Figure 3 reports a measurement, to our knowledge not previously available at this scale, of how PC update mass is distributed across the 4.72×10^6 error cells of the terminal settle. The distribution is strongly concentrated: ranked by magnitude, the top 5% of cells carry $\approx 90\%$ of the squared update mass. The concentration is *rank-form*, and it depends on no tail model: absolute magnitude thresholds do not separate a frontier, since nearly all cells sit below any fixed cutoff, yet ranked by magnitude the mass is extremely concentrated. The rank concentration has a direct engineering consequence for the settle cost discussed in §5: a settle that updates only the top-ranked cells could recover most of the update at a fraction of the work.

As a τ -comparable summary of tail weight we also record a Hill statistic at a fixed depth of the top 5%, $k = 235,929$ of the 4.72M cells: $\alpha = 1.681$ at $\tau=6.4$; the sampling standard error of this fixed-depth statistic, estimated model-free from the empirical dispersion of the log-excesses in the stored histogram, is ≈ 0.002 . Across τ the statistic moves by at most 0.008 (1.673, 1.679, 1.681 at $\tau = 1.0, 3.2, 6.4$), a few sampling errors, so tail shape at fixed depth is nearly τ -invariant. We do not claim a limiting power law: sweeping the tail depth from the top 0.5% to the top 20% moves the Hill statistic monotonically from ≈ 2.2 to ≈ 1.1 with no plateau, several hundred sampling errors of movement (sweep from the same histogram), so α serves as a fixed-depth descriptor of tail weight rather than an estimate of a tail exponent.

Two further questions about this structure matter: where the frontier lives in depth, and whether it needs the teacher. Both now have measured answers. Depth first. The settle activity is spread across the network rather than piled into late blocks: at terminal τ the final block carries the smallest share of the last relaxation step’s update norm, 0.4%. The end-state frontier tells the same story more sharply. Decomposing the top-5% frontier by block, on an exact replication of the recorded settle whose endpoint energy and tail statistics reproduce to

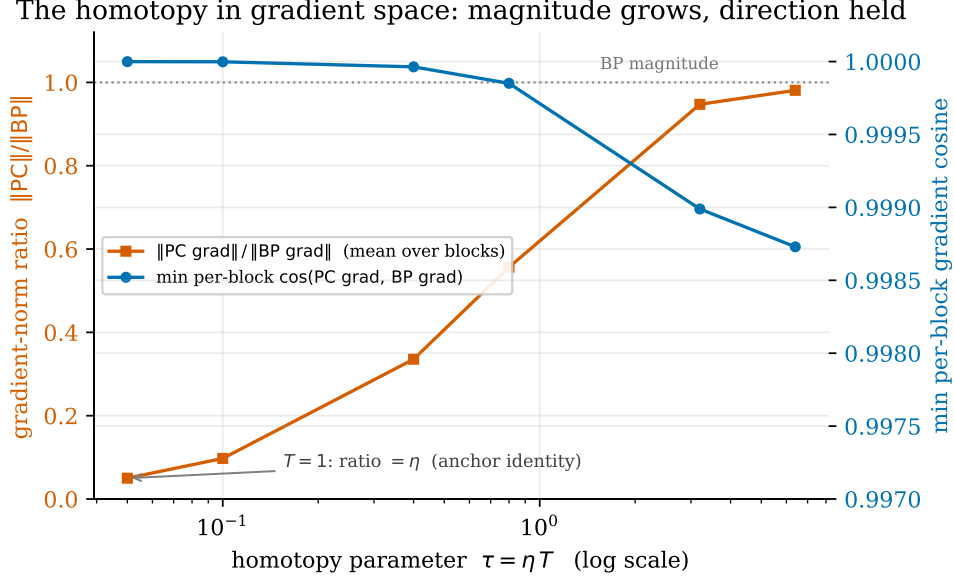


Figure 2: The homotopy in gradient space. Teacher-free probe at each milestone: per-block weight gradients of the settled local objective (3) against plain backpropagation gradients, under a cross-entropy loss on data. The norm ratio (orange, left) starts at the $T=1$ anchor identity $\eta = 0.05$ and grows to 0.98, full backpropagation magnitude; the minimum per-block cosine (blue, right) never falls below 0.998. The learning signal leaves the anchor in magnitude while never leaving backpropagation’s direction.

within 2×10^{-4} relative, blocks 2–5 hold 66% of frontier cells, the final three blocks hold 0.2%, and the share falls away toward the readout: the error mass does not pile into late layers. Second, the teacher. Rerunning the same settle with the distillation term replaced by cross-entropy on data, no teacher present, preserves the structure: the top 5% of cells carry 91.6% of squared update mass, against 90.0% under the training objective, with the same depth profile. Across fresh samples under both objectives the top-5% share stays between 82% and 97% with the same shape, while the fixed-depth Hill statistic ranges from 1.38 to 2.00 across samples and objectives, reinforcing that it is a descriptor of tail weight rather than an exponent.

4 Reproducibility methodology

Acceptance is governed by a pre-registered battery of six criteria (a knowledge-distillation anchor, per-milestone prompt KL, held-out perplexity delta, terminal block cosine, an error-norm-ratio monotonicity condition, and an energy-convergence hold count). During the replicate campaign a methodological issue surfaced that shapes how the planned comparisons are designed. The initial KL and cosine thresholds had been frozen from a *single* pilot run; the four-cell replicate matrix then showed that these thresholds sat *inside* the run-to-run reproducibility noise: a nominal-configuration replicate drifted further from the pilot than a compute-lever variant did. The bars were therefore recalibrated, under a rule fixed before the last replicates were seen, to the strict envelope over the replicate set ($n \geq 3$); and configuration comparisons are henceforth run at *matched seeds* and judged on per-seed deltas, which cancels seed variance. The lesson is general and is carried into the experiments below: never set an absolute acceptance threshold from a single sample of a chaotic training signal.

5 Status and limitations

Three things are established: the PC parameterisation holds a modern transformer at 124M, the learning signal grows from the anchor to full backpropagation magnitude while function is

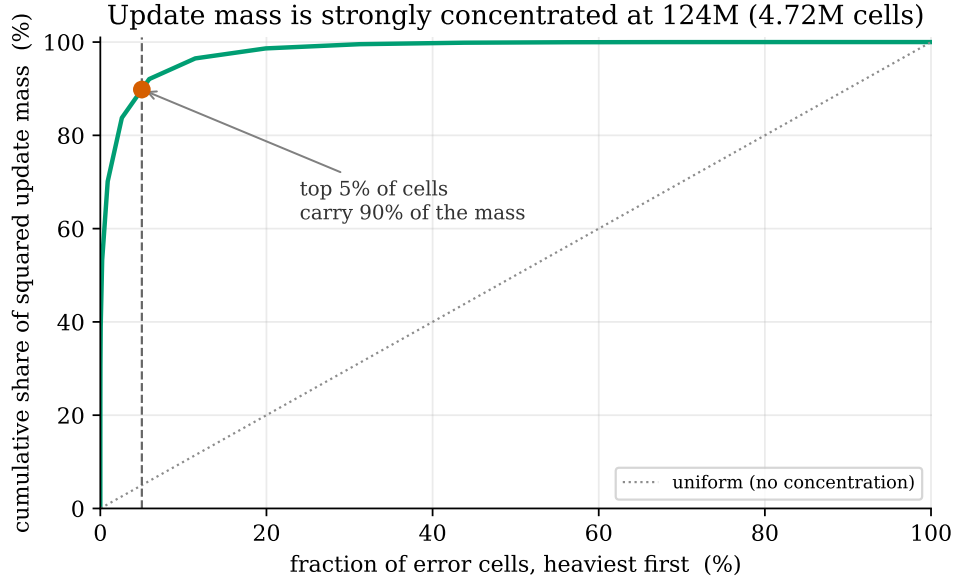


Figure 3: Update-mass concentration at 124M. Cumulative share of squared update mass against the fraction of error cells taken heaviest-first, at the terminal settle ($\tau=6.4$, 4.72M cells). The top 5% of cells carry $\approx 90\%$ of the mass; the dotted line is the no-concentration reference. Hill statistic at the top-5% tail: $\alpha = 1.68$.

preserved, and the error field carries an exploitable rank structure. This is a progress report, so the boundaries are stated with the same force.

- **Re-expression, not from-scratch training.** The teacher did the original learning; the contribution is that PC can *hold* the result at 124M with local updates. Whether PC can *train* a transformer of this size from random initialisation, competitively with backpropagation, is open. The blocker is compute: each PC weight update requires an inner relaxation of T forward/backward passes where backpropagation requires one, so from-scratch PC at scale is one to two orders of magnitude more expensive per update. The concentration measurement of §3.3 is a first lever on exactly this cost.
- **Token scale.** The replicate family uses 5×10^6 tokens and the completed production run 5×10^7 ; both remain far short of pretraining scale.
- **The motivating advantage is not yet demonstrated.** That the PC form forgets less or transfers better than a backpropagation baseline, which is the reason to prefer PC at all, is a designed comparison that has not been run [planned]. Nothing in this report should be read as evidence for it.

6 Companion result: the same transformer inside a symbolic store

The PC question of this report is asked because of a companion result by the author, previously unpublished, which supplies the substrate. The full GPT-2 (124M) forward pass runs natively inside MORK, the MeTTa Optimal Reduction Kernel of the OpenCog Hyperon programme [7, 8]: weights and activations live as expressions in the kernel’s trie-backed space, and the forward pass executes as the kernel’s own rule application over tensor-block expressions, with dense numeric kernels (einsum, attention, normalisation) added by the author as sink operations. The in-store forward matches the reference implementation to a maximum relative logit error of 6.06×10^{-6} with argmax agreement, and a blocked tensor format brings the 12-layer forward from 124.5M store expressions to 7.8M, a $16\times$ reduction, at the identical maximum relative error.

Decoding is incremental and exact. Causal masking means old columns never read new ones, and einsum is bilinear, so appending one token produces exactly a one-column delta through all twelve layers; this is the KV-cache identity derived as incremental view maintenance rather than assumed as a heuristic. A 32-token greedy decode runs in a single kernel invocation, with all 32 tokens equal to the reference oracle.

The two results are halves of one programme. A store-hosted transformer makes weights, activations, and symbolic knowledge one queryable, incrementally maintained structure; what such a substrate cannot do cheaply is a global, synchronised backward pass. Predictive coding is the learning mode that fits it: the updates of §2 are local to each block, and by §3.3 the update mass concentrates on a small ranked frontier of cells, exactly the sparse, incremental write pattern a trie-backed store maintains well. This report establishes the learning-side half at full model scale; the companion result establishes the inference-side half.

7 Planned work

Three lines follow directly. (1) A continual-learning benchmark: sequential-domain training measuring forgetting and transfer for the PC form against backpropagation fine-tuning at matched compute, with the matched-seed, envelope design of §4. (2) A coupled-objective grid separating whether any advantage comes from the PC parameterisation or from objective coupling. (3) Path-faithful settle compression (higher-order integrators at fixed evaluation budget and a rank-frontier settle informed by the measured rank concentration), the enabling technology for pushing toward from-scratch PC training at reduced scale. Results from (1) and (2) are the immediate next milestone. Longer term, the target is the substrate of §6: predictive coding as the learning rule of the store-native transformer, with frontier-compressed settles as the bridge. Figure 4 places the report in this programme.

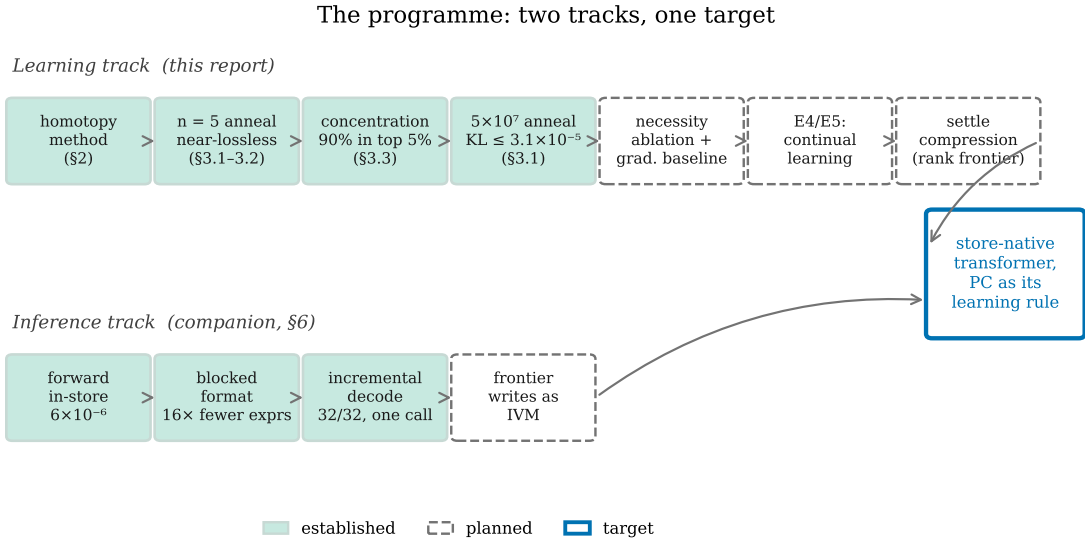


Figure 4: Programme roadmap. Green boxes are established results (this report and the companion result of §6); dashed boxes are planned (§7). The tracks converge on a store-native transformer whose learning rule is predictive coding over the measured rank frontier.

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